Group Assignment Cover Sheet

MBA 2024/2025

Note: This cover sheet should be incorporated within your assignment document and should not be submitted as a separate document.

Group Blind Grade numbers (BGNs) :

6077I
5935N
6080L
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6171L

Group No: 10

Assignment Code: MBA58

Name of Course: New Venture Finance

Lecturer for this Coursework: Robert Wardrop

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Each group should only submit ONE copy of the assignment, but all group members are responsible for ensuring that the assignment is submitted on time.

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1a.

Alpha

- Pre-money Valuation (£4.25 per share): x £4.25 = £17,899,309
- Post-money Valuation MTA (£0.01 per newly issued share): £17,899,309 + £400,000 = £18,299,309

Beta

- Pre-Money Valuation (£0.90 per share): 4,636,386 × £0.90 = £4,172,747.4
- Post-Money Valuation: £4,172,747.4 + £800,000 = £4,972,747.40

Gamma

- Pre-money Valuation (£2.80 per share): 2,654,807 x £2.80 = £7,433,460
- Post-money Valuation (£0.37 per newly issued share): £7,433,460 + £600,000 = £8,033,460

1b.

Alpha

Assumptions:

1. Exit in 5 years, assuming IPO or been entirely purchased.
2. Exit Value in 5 year based on energy sector:

	Projected Financials for Year 5	Industry Valuation Multiples	Estimated Exit Valuations
Revenue:	£27.34 million	4.0x	£109.36M
EBITDA:	£12.8 million	9.0x	£115.2M

3. Discount rate: 20% for mid size energy company.
4. £3M fundraising before exit
5. Owners in Cap Table: N, PV & R.

6. Timing of future fundraising

1st round(Year 1)
£400k

Step 1: Discounted Factor

Discounted Terminal Value= Terminal Value/$(1 + r)^n$ = £115m/$(1+ 0.20)^5$ = £46.22M

Step 2: Ownership dilution given 2.19% equity transferred to Angel Investors:

		Pre Financing		Post Financing	
Shareholder ID	Description/Name	Shares	%	Shares	%
1	N	465218	11.05%	465218	0.108046
2	PV	423146	10.05%	423146	0.098275
3	R	423146	10.05%	423146	0.098275
Total Ownership			31.14%		
Ownership Post Funding	= 31.14%*(1-2.19%)	=	30.46%		
Future Dilution	=1-(30.46%/31.14%)	=	2.18%		

Step 3: Reduce pre-seed valuation estimate

Present Value of company today	£ 46,220,000.00
Future dilution	0.0218
Implied valuation (PV*(1-dilution))	£ 45,212,404.00

Estimated valuation for Alpha is **£45.21M.**

Beta

Assumptions

1. Exit in 5 years
   - In the medical device sector, the median time to exit is approximately 7 years
   - With first close of funding that help achieving profitability can make BETA more attractive to potential acquirers or investors, potentially shortening the time to exit.

2. Exit Value: £50M

	Projected Financials for Year 5	Industry Valuation Multiples	Estimated Exit Valuations
Revenue:	£12.2 million	4.5x	£54.9million
EBITDA:	£4.7 million	10.2x	£47.94million

3. Discount rate: 30%
   - Generating £750K revenue and serving 76 global customers, proving market traction
   - Acquiring 3 new hospital clients per month, indicating predictable adoption and scalability
   - Seeking investment for growth and profitability, rather than early-stage R&D

- Industry benchmarks suggest 30-40% discount rates for Mid-Stage MedTech and BETA's revenue and adoption trends align with the lower end of this range

4. £3M fundraising before exit

5. Timing of future fundraising

1st round (Year 1)	2nd round(Year 2)	3rd round (Yea2)
£800k	£1M	£1.2M

Step 1: Discounted Terminal Value
The Discounted Terminal Value = Terminal Value/$(1 + r)^n$
The Discounted Terminal Value = £50m/$(1+ 0.30)^5$
The Discounted Terminal Value = £13.47 million

Step 2: Ownership dilution resulting from future funding rounds

	Financing Round		Valuation of Round		Realised Value at Exit		Years to Exit	IRR	Return Multiple
	Amount	Ownership	Pre Money	Post Money	Ownership	Value			
Founders	0	n/a	0	n/a	64.9%	$ 32,450,000.00		n/a	n/a
Seed	£ 800,000.00	18.4%	£ 4,172,747.40	£ 4,972,747.40	11.3%	$ 5,649,995.00	4.0	63%	7.1
Series A	£ 1,000,000.00	15.0%	£ 5,650,886.00	£ 6,650,886.00	10.8%	$ 5,418,750.00	3.0	76%	5.4
Series B	£ 1,200,000.00	15.3%	£ 6,650,886.00	£ 7,850,886.00	13.0%	$ 6,500,000.00	3.0	76%	5.4
Sum	$ 3,000,000.00				100.0%	$ 50,000,000.00			16.7

Future dilution = 1-(11.3%/18.4%)=38.6%

Step 3: Reduce pre-seed valuation estimate

Present Value of company today	£ 13,470,000.00
Future dilution	0.386
Implied valuation	£ 8,272,329.64

Estimated valuation for Beta is **£8.3 million**

Gamma

Assumptions

1. Exit in 5 years
2. Annual Growth Rate for 2018-2023 based on 2012 to 2017): £(900/750m)^(1/5) -1 = 3.7%
3. Total UK market in 5 years = £900m*(1+3.7%)^5 = £1.3bn
4. As no single company has a more than 15% market share, there is no compelling monopoly power. We assume Gamma's market share is ~1-3%
5. Exit Multiple – 3x Earnings
6. Target rate of return : Average Venture Capital Discount rate 45%

7. Future dilution capped at 30%

Step 1: Estimated Market Share in 5 years

- Worst case: Gamma captures 1% market share
- Base case: Gamma captures 2% market share
- Best case: Gamma captures 3% market share

Market Share	Revenue at Exit (£B)	Revenue (£M)
1%	1.3B × 1% = 0.013B	£13M
2%	1.3B × 2% = 0.026B	£26M
3%	1.3B × 3% = 0.039B	£39M

Step 2: Exit Muliplier to estimate Terminal Value at exit

Market Share	Revenue (£M)	Exit Multiple	Exit Value (£M)
1%	£13M	3.0x	£39M
2%	£26M	3.0x	£78M
3%	£39M	3.0x	£117M

Step 3: Using 45% as discount rate to calculate PV today

Market Share	Exit Value in 5 years (£M)	Present Value (£M) (Before Dilution)
1%	£39M	£7.4M
2%	£78M	£14.8M
3%	£117M	£22.2M

Step 4: Future dilutive effect capped at 30%

Current dilutive effect to owners resulted from £600,000 raised:

- Pre-money ownership: 1,885,693 / 2654807 = 71%
- Post-money ownership: 1,885,693 /4,276,429 = 44%
- Dilutive effect: 1−71.04%/44.10%=38.0%

If we assume another round where new shares issued also dilute existing shares by ~38%.

- New shares issued (similar effect): 4,276,429×62%=2,612,728
- Total shares after second round: 4,276,429+2,612,728=6,889,157
- Ownership Post-Second Round=1,885,693/6,889,157=27.4%

With an expected market share at 1% and a diluted ownership at 27.4%, Gamma can exit at £39M in 5 years, the original owners will get £10.7M with PV today at £2.03M. If the company does nothing and exit today, the owners can get £8,033,460*44.1% = £3.5M. Unless the fundraising supports strong market penetration, we assume

Gamma will not have aggressive fundraising to avoid excessive ownership dilution. We cap future dilution at 30%.

Step 5: Reduce pre-seed valuation estimate

Under the VC method implied valuation, base case scenerio would be:

- Pre-dilution company value= £14.8M
- Dilutive effect: 30%
- Post-dilution company value= £14.8Mx (1-30%) = £10.36M
- Return to angel investors in five years: £14.8M*38%*(1-30%)/£8M*38%=29.5% < 45% target rate

2.

Future Dilution Scenarios

Company	Chances for Future Rounds	Dilution Risk
Alpha	Moderate - Strong government support & revenue potential	Low - Moderate
Beta	High - Needs more capital to scale from smaller valuation	High
Gamma	High - Already significant dilution in this round	Very High

Assuming a 5-year exit strategy with below potential exits at 3x, 5x, and 10x multiples for each company.

Company	Post-Money Valuation (£)	Ownership Stake (%)	3x Exit (£)	5x Exit (£)	10x Exit (£)
Alpha	18.3M	2.19%	*£1.2M	*£2.0M	*£4.0M
Beta	4.97M	16%	£2.4M	£4.0M	£8.0M
Gamma	8.03M	18%	£4.3M	£7.2M	£14.4M

** Note on Alpha: To compare it against Beta and Gamma, it is assumed that as mature company they can have an exit via an IPO or are entirely purchased by another company.*

These are Insights on Expected Returns

- Gamma offers the highest potential ROI, but its high dilution risk means actual returns could be much lower.
- Beta offers a strong upside, with high ownership and exit potential, but also has high dilution risk.
- Alpha is the safest investment, offering modest returns with lower dilution risk and stronger industry backing.

Final Investment Recommendation

We recommend a balanced strategy in investing in each of the company: £60K in Alpha, £50K in Beta, £40K in Gamma offers a mix of stability and upside. This approach leverages Alpha's stability and proven traction, Beta's high growth potential, and Gamma's moderate valuation with scaling potential.

Investing £60,000 in Alpha ensures exposure to a well-established company with global industry partnerships and strong government backing, while minimizing dilution risk. Although Alpha's valuation is higher, it offers long-term stability and lower downside risk compared to the other two companies as a relatively matured company.

Increasing the investment in Beta to £50,000 takes advantage of its lower pre-money valuation, offering a larger ownership stake and higher potential upside. However, Beta carries a higher dilution risk due to its funding needs, meaning future rounds could impact investor returns.

Keeping £40,000 in Gamma provides moderate exposure to a company with solid growth prospects, while acknowledging its significant dilution risk. With a well-balanced allocation, this strategy allows for both high-return opportunities and downside protection, making it an optimal mix for a diversified investment portfolio.